

Linux Task 03_anju: Process Management, System Monitoring & Basic Shell Scripting

Objective The purpose of this task is to understand how Linux manages processes, monitors system resources, and automates tasks using shell scripts. These are essential skills for Linux Administrators, SOC Analysts, and Cyber Security Professionals. Part A: Process Monitoring

Part A: Process Monitoring

- ps
- ps aux
- top
- htop
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1. What is a Process?

A process is a program that is currently running on the system. Every application or command executed in Linux runs as a process. Linux manages processes to ensure efficient use of system resources.

2. What is a PID?

PID stands for Process ID. It is a unique number assigned to every running process. Linux uses the PID to identify and manage processes.

3. Which process is consuming the most CPU?

The process consuming the most CPU is redis.

- PID: 163
- Process Name: redis
- CPU Usage: approximately 0.7%

4. Which process is consuming the most Memory?

the process using the most memory is root process with PID 300.

- **PID:** 300
- **Memory Usage (RES):** 32240 KB
- **%MEM:** approximately 0.4%

Part B: Process Management

Item	Value
Process Name	sleep
PID Found	2123
Command Used	kill 2123
Result	Process terminated successfully

Part C: System Monitoring

- free-h
- df-h
- uptime
- uname -a

Information	Output
Total RAM	7.4 GiB
Available RAM	6.8 GiB

Disk Usage	/dev/sdd: 1007G total, 2.1G used, 954G available (1% used)
System Uptime	3:08 (3 hours 8 minutes)
Kernel Version	Linux LAPTOP-P6ICEUJP 6.6.114.1-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Mon Dec 1 20:46:23 UTC 2025 x86_64 GNU/Linux

Part D: Service Monitoring

- `systemctl status ssh`
- `systemctl status NetworkManager`

1. What is a Service?

A service is a background program that runs continuously to provide specific functions such as networking, logging, or web hosting.

2. Why are services important?

Services keep essential system functions running automatically. They ensure that applications and system components work properly.

3. How can a stopped service affect a system?

If an important service stops, related functionality may fail. For example, if the network service stops, internet connectivity may be lost.